

Name:

Partner Check:

BUID:

Workshop 1: Matrix-Vector Equations

1. *Review.* Determine if the vector $\begin{bmatrix} 9 \\ 3 \\ -14 \end{bmatrix}$ is in the span of $\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix} \right\}$.

2. Calculate the matrix-vector multiplication $\begin{bmatrix} 1 & -4 & 3 \\ -6 & 2 & 5 \end{bmatrix} \begin{bmatrix} -5 \\ -1 \\ -1 \end{bmatrix}$.

3. Write a vector equation and a linear system with the same solution set as the matrix-equation $\begin{bmatrix} -4 & 5 & 5 \\ 0 & 1 & 1 \\ -9 & 3 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$.

4. Determine if the equation $\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ -2 & -1 & -4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 9 \\ 3 \\ -14 \end{bmatrix}$ has a solution. If it does, then determine a general form solution.

5. Determine if the equation $\begin{bmatrix} 1 & 3 & -1 \\ 1 & -1 & 3 \\ -1 & 2 & -4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 10 \\ -2 \\ 5 \end{bmatrix}$ has a solution. If it does, then determine a general form solution.
6. Determine the values of b_1 and b_2 such that the RREF of $\begin{bmatrix} 1 & -1 & b_1 \\ -2 & 2 & b_2 \end{bmatrix}$ does not have the row $[0 \ 0 \ 1]$.
7. Determine the values of b_1 , b_2 , and b_3 such that the RREF of $\begin{bmatrix} 1 & -1 & b_1 \\ 2 & -1 & b_2 \\ 0 & -1 & b_3 \end{bmatrix}$ does not have the row $[0 \ 0 \ 1]$.
8. Determine if $\begin{bmatrix} -4 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} 8 \\ -10 \end{bmatrix}$ span all of $\mathbb{R}^2$. Also determine if $\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 19 \\ 11 \\ 0 \end{bmatrix}$ span all of $\mathbb{R}^3$. Explain your answers.
9. Determine a matrix $A \in \mathbb{R}^{2 \times 3}$ with distinct entries such that the equation $A\mathbf{x} = \mathbf{0}$ has $\mathbf{1}$ (the all-ones vector) as a solution.